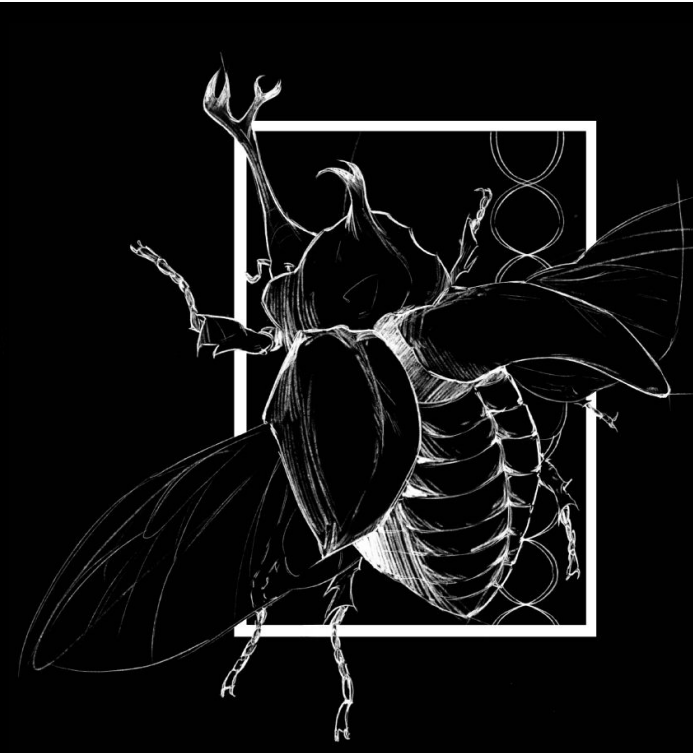




Background

Passifloraceae is a clade of flowering plants that includes shrubs and trees. Studies have shown that there is a vast variation in genome sizes between each species in the clade Passifloraceae, with species ranging from ~200 Mp to as high as ~5000 Mp. Each species also has varying haploid numbers due to dysploidy (single chromosomal changes) or polyploidy (whole genome duplication) events. This research project investigates how the haploid number of each species correlates with its genome size. I hypothesize that the haploid number of Passifloraceae species may correlate with the species genome sizes, and a higher haploid number correlates with a larger genome size due to an abundance of long-terminal repeats (LTR) and insertion sequences (IS). To test this hypothesis, I collected data on the species genome sizes and haploid values using a PGLS model consisting of a Brownian motion correlation structure, utilizing a sample size of 56 species. In this study, I found that larger haploid numbers are strongly correlated with larger genome sizes, and smaller haploid numbers are strongly correlated with smaller genome sizes. Based on these findings, I can conclude that the haploid number provides a strong positive correlation with genome size.

Study Design & AI

This study utilized a literature review process to identify the genome sizes of each species studied, and utilized the provided supplemental materials from the professor that consisted of the haploid numbers of these species. Through literature review and the provided supplemental data, 56 species were analyzed with haploid numbers ranging from 5 to 20 and genome sizes ranging from ~200 Mp to 5500 Mp. The genome sizes and haploid numbers were then combined into a CSV file that indicated the species name, the haploid number, and its genome size. To further analyze this study, an R code was utilized using a PGLS model consisting of a Brownian motion correlation structure to account for phylogenetic relatedness. This code was created using supplemental materials and the CSV file, which generated a residual plot that provided a visual representation of the model's fit in predicting genome size based on the haploid number. In addition, AI materials such as Claude and ChatGPT were utilized for guidance in developing an R code and generating the figure creating the scatterplot.

Model Fitting

The model used for this experiment was a PGLS model since these species share evolutionary relationships and are therefore not independent of each other. This experiment also utilized a Brownian motion correlation structure, which assumes that trait evolution occurs randomly and gradually across an evolutionary time period. The haploid number was the predictor variable (x), and the genome size was the response variable (y). The regression was run in R Studio using the package nlme and the gls() function using the corBrownian function from the package ape. There was also a phylogenetic tree incorporated into the model that accounted for the relatedness across 56 different species.

Results

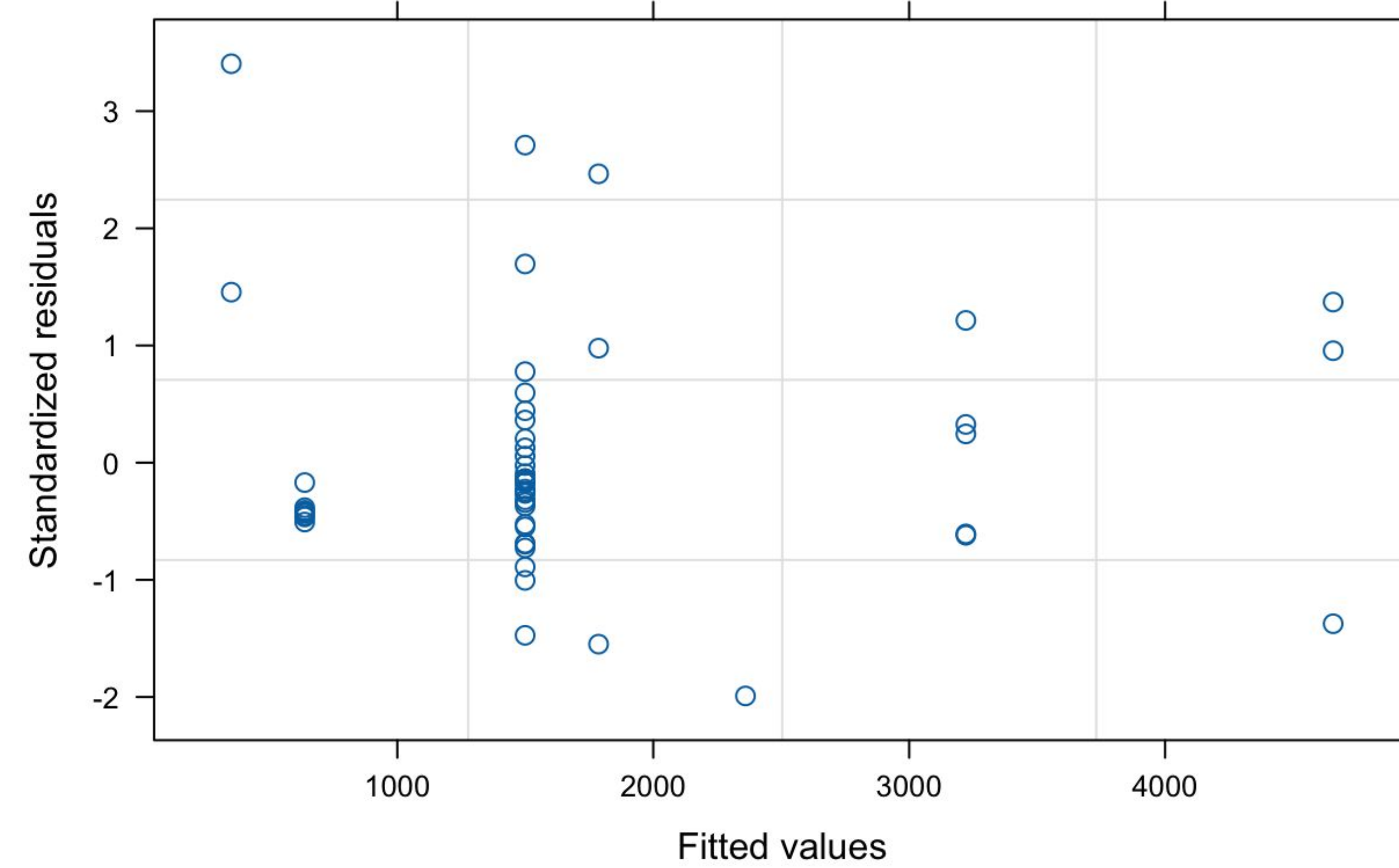


Figure 1. Residual plot of the haploid number versus the genome sizes of each species

This model used a PGLS plot, with the fitted values predicting genome size from the haploid number. The standardized residuals represent the differences between the predicted genome and the actual genome sizes. In this model, heteroscedasticity is absent and illustrates an even scatter of residual points with a p-value of less than 0.001. This indicates that the haploid number is a significant predictor of genome size.

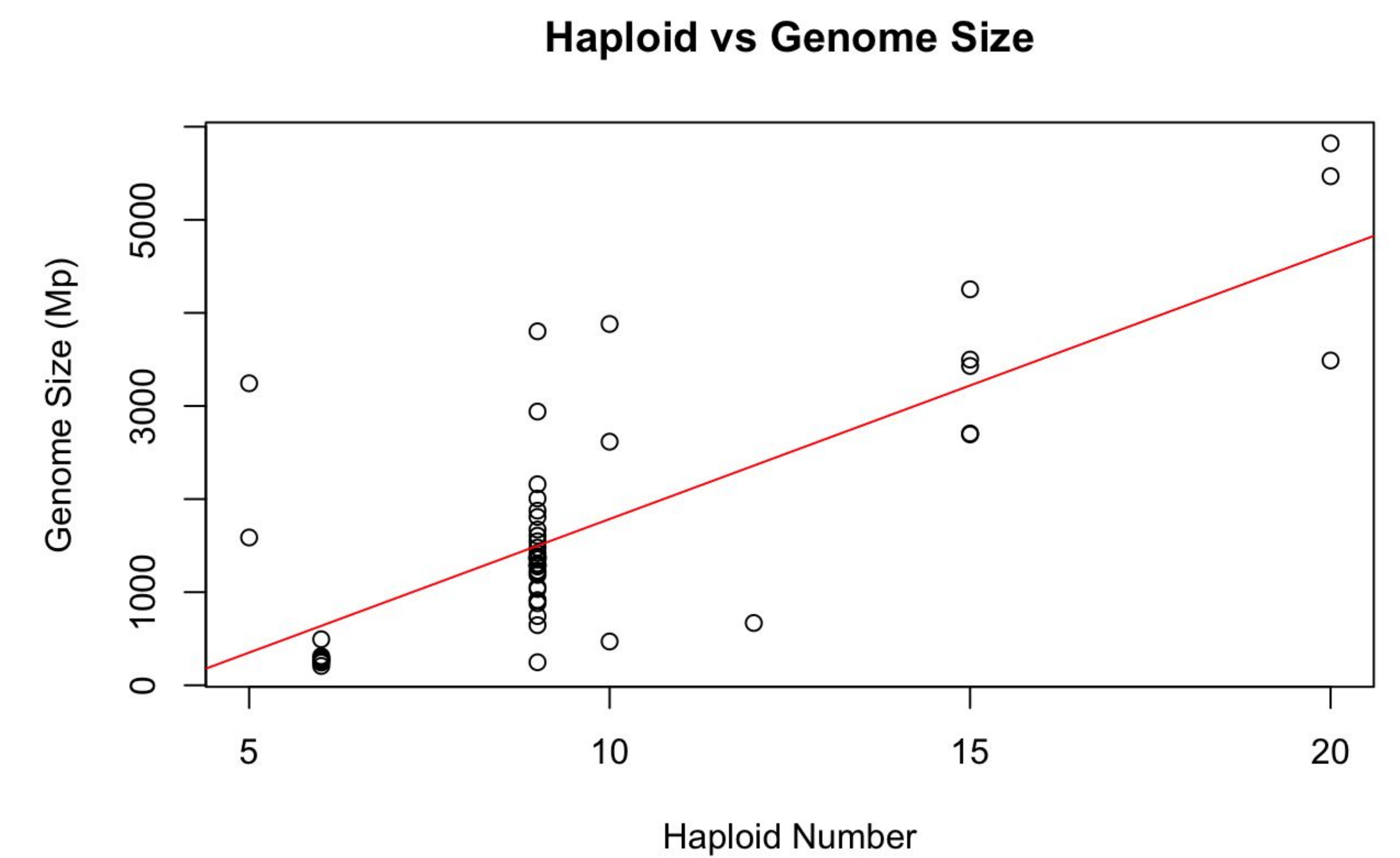


Figure 2. Scatter plot between haploid number and genome size (Mp)

This model utilizes a scatter plot to plot the correlation between the haploid number and genome size (Mp). The red line represents the regression line predicting the relationship between the haploid number and genome size. The correlation coefficient using Pearson's model is 0.7568716, indicating a positive correlation, confirming that the haploid number is a significant predictor of genome size.



Discussion

I found that with a p-value less than 0.001, Figure 1 portrays that the haploid number is a strong predictor of the genome size in the species of Passifloraceae. The data supported my earlier prediction that a lower haploid number contributes to a smaller genome size and a larger haploid number contributes to a larger genome size. This can be supported by the Planta (2021) paper, which suggests that genome size variations are driven by LTR retrotransposons. In addition to retrotransposons, genome size variation is also a result of dysploidy and polyploidy, which, according to the Mota and Loureiro (2016) paper, are the main drivers for Passifloraceae diversification. The results also showcase outlier points where species showcased much larger and smaller genomes than what was predicted. These outliers are best explained by insertion sequences, or transposable elements, as these can cause variations in genome size. Some of the limitations present in this research are the sample size, which consisted of only 56 species. For conducting further research, the sample size can be expanded to include more diversity in haploid numbers and genome sizes.

Conclusions & Acknowledgements

These results support that the haploid number is a valuable predictor of genome size. This finding is significant in understanding how transposable elements, insertion sequences, and ploidy play a significant role in the diversity of genome sizes. These findings suggest that insertion sequences, transposable elements, and ploidy events, along with chromosomal changes, are significant mechanisms that contribute to the diversity of genomes in this clade. Further research on this clade should expand the sample size to bring a broader range of haploid numbers, genome sizes, and additional genera to further encompass the Passifloraceae clade and further characterize the evolutionary patterns in this clade. I would like to thank my professor, Dr. Blackmon, for their assistance, guidance, and support through this project. I would also like to thank their peers for providing support and assistance, and Dr. Lillvis for helping run the CUREs program and making this a great learning experience.